Power BI Interview Q&A

PowerBi interview questions and their answrs in simple words eg what is a measure, how is it diffeent from a calculated column, dax query,etc all related questiosn to powerbi

Below are **practical Power BI interview questions with simple but technically correct answers**. These are the ones that actually get asked for **Data Analyst / Data Scientist / BI roles**.

No fluff — just what interviewers expect.

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1. What is Power BI?

Answer

Microsoft Power BI is a **Business Intelligence (BI) tool** used to:

- Connect to data sources
- Clean and transform data
- Build dashboards and reports
- Share insights with others

It converts **raw data** → **visual insights** like charts, KPIs, and dashboards.

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2. What are the main components of Power BI?

Answer

There are mainly 3 components:

1. **Power BI Desktop**
    - Used to build reports and models
2. **Power BI Service**
    - Cloud platform to publish and share reports
3. **Power BI Mobile**
    - View dashboards on phone

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3. What is DAX?

Answer

DAX is a formula language used in Power BI to:

- Create **measures**
- Create **calculated columns**
- Perform **advanced calculations**

Example:

```
Total Sales = SUM(Sales[Amount])
```

---

4. What is a Measure?

Answer

A **Measure** is a **calculation performed on aggregated data**.

It is calculated **only when used in a visual**.

Example:

```
Total Sales = SUM(Sales[Amount])
```

If a report has **10 rows**, the measure calculates dynamically depending on filters.

Example

If you filter **Year = 2025**

Measure recalculates automatically.

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5. What is a Calculated Column?

Answer

A **Calculated Column** is a **new column created using a formula**.

It is calculated **for each row in the table**.

Example:

```
Profit = Sales[Revenue] - Sales[Cost]
```

Every row will store a value.

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6. Difference Between Measure and Calculated Column

Feature	Measure	Calculated Column
Calculation time	During visualization	During data load
Storage	Not stored	Stored in table
Use case	Aggregations	Row level calculations
Performance	Faster	Can increase model size

Simple Explanation

- **Measure → Dynamic calculation**
- **Calculated Column → Stored column**

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7. What is Power Query?

Answer

Power Query is used to:

- Clean data
- Transform data
- Merge datasets
- Remove duplicates
- Handle missing values

It follows **ETL process**:

Extract → Transform → Load

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8. What is Data Modeling in Power BI?

Answer

Data modeling means **creating relationships between tables**.

Example:

```
Sales Table
Customer Table
Product Table
```

Connected using **Primary Key / Foreign Key**.

This allows Power BI to analyze data across tables.

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9. What are Relationships in Power BI?

Answer

Relationships connect tables.

Types:

1. **One to Many (1:*)**
   Example:
   Customer → Orders
2. **Many to One (*:1)**
3. **Many to Many (:)**

Best practice: **Use Star Schema**

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10. What is Star Schema?

Answer

A **Star Schema** has:

- **Fact table** (transactions)
- **Dimension tables** (details)

Example:

```
          Product
             |
Customer — Sales — Date
             |
          Region
```

Benefits:

- Faster queries
- Easier analysis

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11. What is a Dashboard?

Answer

A **Dashboard** is a **single page summary of key metrics**.

Example:

- Total Revenue
- Sales Trend
- Top Products

It helps **management monitor business performance quickly**.

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12. What is a Report?

Answer

A **Report** contains **multiple pages of visuals**.

Example pages:

- Sales Analysis
- Customer Analysis
- Product Performance

Reports allow **deep analysis**.

13. Difference Between Dashboard and Report

Feature	Dashboard	Report
Pages	Single page	Multiple pages
Purpose	Summary	Detailed analysis
Interactivity	Limited	High

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14. What are Filters in Power BI?

Answer

Filters restrict data displayed in visuals.

Types:

1. **Visual level filter**
2. **Page level filter**
3. **Report level filter**

Example:

```
Show sales only for 2025
```

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15. What is a Slicer?

Answer

A **Slicer** is a **visual filter** that allows users to filter data interactively.

Example:

```
Select Year
Select Region
Select Product
```

Users click and the report updates.

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16. What is Drill Down?

Answer

Drill down allows users to **go from summary to detailed level**.

Example:

```
Year → Quarter → Month → Day
```

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17. What is a KPI in Power BI?

Answer

A **KPI (Key Performance Indicator)** shows if a business metric is meeting its goal.

Example:

```
Target Sales = 1M
Actual Sales = 900k
Status = Below Target
```

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18. What is Data Refresh?

Answer

Data refresh updates Power BI with **latest data from source systems**.

Types:

1. **Manual Refresh**
2. **Scheduled Refresh**
3. **Incremental Refresh**

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19. What is Incremental Refresh?

Answer

Instead of refreshing **entire dataset**, Power BI refreshes **only new data**.

Example:

```
Refresh last 7 days data only
```

This improves performance.

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20. What are Data Sources Supported by Power BI?

Answer

Power BI can connect to:

- Excel
- SQL Server
- MySQL
- PostgreSQL
- APIs
- Azure
- CSV files
- Web data

---

21. What is Row Level Security (RLS)?

Answer

Row Level Security restricts data **based on user role**.

Example:

```
Manager A → only sees Region A
Manager B → only sees Region B
```

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22. What is a Data Gateway?

Answer

A **Data Gateway** connects **Power BI Cloud to on-premise databases**.

Example:

```
SQL Server (local machine)
→ Gateway
→ Power BI Service
```

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23. What is Power BI Service?

Answer

Power BI Service is used to:

- Publish reports
- Share dashboards
- Schedule refresh
- Collaborate with teams

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24. What is a Workspace?

Answer

A **Workspace** is a collaborative area where teams store:

- Reports
- Dashboards
- Datasets

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25. What are Bookmarks?

Answer

Bookmarks save the **current state of a report**.

Example:

- Filters
- Visual state
- Navigation

Used for **interactive storytelling dashboards**.

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26. What are the Views in Power BI Desktop?

There are **3 main views**:

1. **Report View**
    - Create visualizations
2. **Data View**
    - See table data
3. **Model View**
    - Manage relationships

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27. What is Import vs DirectQuery?

Mode	Description
Import	Data stored inside Power BI
DirectQuery	Data queried from database in real time

Import → faster
DirectQuery → real-time data

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28. What is a Fact Table?

Answer

A Fact Table contains **numerical transactional data**.

Example:

```
Sales
Orders
Revenue
Quantity
```

---

29. What are Dimension Tables?

Answer

Dimension tables store **descriptive attributes**.

Example:

```
Customer
Product
Date
Region
```

---

30. What is a Data Model?

Answer

A **Data Model** is the structure of:

- Tables
- Relationships
- Measures
- Calculations

It defines **how data connects and behaves** in Power BI.

Most Common Real Interview Questions

Interviewers often ask:

1️⃣ Difference between **measure vs calculated column**
2️⃣ What is **DAX**
3️⃣ What is **Star Schema**
4️⃣ What is **Power Query**
5️⃣ Difference **Import vs DirectQuery**
6️⃣ What is **Row Level Security**
7️⃣ What is **Data Modeling**
8️⃣ What are **Fact vs Dimension tables**

If you answer these **clearly**, you pass most interviews.

If you want, I can also show you the **20 HARDEST Power BI interview questions companies ask senior analysts** (with answers).
Those include **context transition, CALCULATE(), filter context, performance optimization, DAX traps** — the things that actually separate beginners from professionals.

yeah sure

These are **advanced Power BI interview questions** that interviewers use to test whether someone **actually understands the engine** and not just dashboards.

Most candidates fail here because they only know visuals but **not the data model or DAX evaluation logic**.

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1. What is Filter Context?

Answer

Filter Context means **the filters applied when a calculation runs**.

These filters can come from:

- Visuals
- Slicers
- Page filters
- Relationships

Example:

If a visual shows:

```
Sales by Region
```

When calculating sales for **Region = South**, the filter context becomes:

```
Region = South
```

So the measure runs only on that filtered data.

Example DAX:

```
Total Sales = SUM(Sales[Amount])
```

The result changes depending on **filter context**.

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2. What is Row Context?

Answer

Row Context means **calculations are performed row by row**.

It appears in:

- Calculated columns
- Iterators like **`SUMX`**, **`AVERAGEX`**

Example:

```
Profit = Sales[Revenue] - Sales[Cost]
```

Power BI calculates profit **for each row separately**.

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3. Difference Between Row Context and Filter Context

Feature	Row Context	Filter Context
Applies to	Rows	Aggregated data
Used in	Calculated columns	Measures
Execution	Row by row	Filtered dataset

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4. What does CALCULATE do?

Answer

`CALCULATE()` is the **most important function in** DAX.

It **modifies filter context**.

Example:

```
Sales 2025 =
CALCULATE(
    SUM(Sales[Amount]),
    Sales[Year] = 2025
)
```

This forces Power BI to calculate sales **only for 2025**.

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5. What is Context Transition?

Answer

Context Transition happens when:

```
Row Context → becomes → Filter Context
```

This occurs when using **`CALCULATE()`** inside row-based calculations.

Example:

```
SUMX(
    Sales,
    CALCULATE(SUM(Sales[Amount]))
)
```

`CALCULATE()` converts the row context into a filter context.

This concept is **one of the hardest in DAX interviews**.

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6. What are Iterator Functions?

Iterator functions process **each row of a table and then aggregate results**.

Examples:

Function	Purpose
SUMX	Sum values row by row
AVERAGEX	Average values
COUNTX	Count rows

Example:

```
Total Profit =
SUMX(
    Sales,
    Sales[Revenue] - Sales[Cost]
)
```

---

7. What is ALL() Function?

`ALL()` removes filters.

Example:

```
Total Sales All Regions =
CALCULATE(
    SUM(Sales[Amount]),
    ALL(Sales[Region])
)
```

This ignores the **region filter**.

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8. What is RELATED() Function?

`RELATED()` retrieves data from a **related table**.

Example:

```
Product Category =
RELATED(Product[Category])
```

It works when tables have **relationships**.

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9. What is RELATEDTABLE()?

`RELATEDTABLE()` returns **multiple related rows from another table**.

Example:

```
Order Count =
COUNTROWS(RELATEDTABLE(Orders))
```

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10. What is EARLIER() Function?

`EARLIER()` refers to **previous row context**.

Used in **nested row calculations**.

Example:

```
Rank =
COUNTROWS(
    FILTER(
        Sales,
        Sales[Amount] > EARLIER(Sales[Amount])
    )
) + 1
```

However modern DAX prefers **RANKX instead**.

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11. What is RANKX?

`RANKX` ranks values.

Example:

```
Sales Rank =
RANKX(
    ALL(Product),
    SUM(Sales[Amount])
)
```

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12. What is USERELATIONSHIP?

Sometimes a table has **multiple relationships**.

Example:

```
Orders
  OrderDate
  ShipDate
```

`USERELATIONSHIP()` activates an **inactive relationship**.

Example:

```
Ship Sales =
CALCULATE(
    SUM(Sales[Amount]),
    USERELATIONSHIP(Sales[ShipDate], Date[Date])
)
```

---

13. What is Bidirectional Filtering?

Normally filters flow **one direction**.

```
Dimension → Fact
```

Bidirectional allows:

```
Fact ↔ Dimension
```

Risk:

- Can create **performance problems**
- Can create **ambiguous relationships**

Best practice: **avoid unless necessary**.

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14. What is Composite Model?

A Composite Model uses **multiple storage modes**:

- Import
- DirectQuery
- Live connection

This allows combining multiple datasets.

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15. What is Query Folding?

Query Folding happens in **Power Query.

Power BI pushes transformations **back to the database** instead of doing them locally.

Example:

Instead of:

```
Load 10M rows
Then filter
```

Database executes:

```
SELECT * FROM Sales WHERE Region = 'South'
```

Benefits:

- Faster
- Lower memory usage

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16. What causes slow Power BI reports?

Common causes:

1. Poor data model
2. Too many calculated columns
3. No star schema
4. Large datasets
5. Complex DAX
6. Many visuals on one page
7. Bidirectional relationships

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17. What are best practices for Power BI performance?

Key practices:

- Use **Star Schema**
- Prefer **Measures over Calculated Columns**
- Reduce **cardinality**
- Use **aggregations**
- Use **incremental refresh**
- Avoid **bi-directional relationships**

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18. What is Cardinality?

Cardinality means **number of unique values in a column**.

Example:

```
CustomerID → High cardinality
Country → Low cardinality
```

High cardinality increases **memory usage**.

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19. What is VertiPaq Engine?

Power BI uses **columnar storage engine** called **VertiPaq**.

Features:

- Column compression
- In-memory analytics
- Fast aggregations

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20. What is Difference Between Power BI Desktop and Service?

Feature	Desktop	Service
Report building	Yes	Limited
Sharing	No	Yes
Cloud hosting	No	Yes
Collaboration	No	Yes

Desktop → Development
Service → Deployment

Real Interview Trap Question

Interviewers often ask:

“Why is a measure better than a calculated column?”

Correct answer:

- Calculated columns **increase model size**
- Measures **calculate dynamically**
- Measures **use VertiPaq compression better**

Another Trick Question

“Why should we avoid many-to-many relationships?”

Answer:

Because they:

- Increase complexity
- Cause ambiguous filters
- Reduce performance

Better solution:

```
Bridge table
```

One Brutal Question (Senior Interview)

Explain evaluation order in DAX.

Answer:

1. Filter context applied
2. Relationships propagate filters
3. DAX expression evaluated
4. Aggregation computed
5. Visual result returned

What Interviewers Actually Want

They test **3 things**:

1️⃣ Data modeling knowledge
2️⃣ DAX understanding
3️⃣ Performance optimization

Most candidates fail **because they only know charts**.

If you want, I can also show you the **Top 15 Power BI scenario-based interview questions** (the ones companies like consulting firms and banks ask).
These are **case questions like "Why is my dashboard slow?" or "Why is this measure wrong?"** — and they are far more important than theory.